CSE331: Automata and Computability

Assignment 1 | Total: 60 Marks

Deadline: Day of Quiz 2

1. **[10 marks]** Convert the following regular languages to their equivalent DFAs

(a) L1 = { w ∣ the number of 0's in w is a multiple of three }, Σ = {0, 1}*
(b) L2 = { w ∣ the third symbol from the right in w is 1 }, Σ = {0, 1}*
(c) L5 = {w ∣ the third symbol from the left is 1 and the last symbol is 0}, Σ = {0, 1}*
(d) L4 = {w ∣ the number of 1's in w is even and the number of 0's is odd }, Σ = {0, 1}*
(e) L5 = {w ∣ w contains the substring 10^m1 where m >= 0 }, Σ = {0, 1}*

2. **[10 Marks]** The symmetric difference of the languages L1 and L2, denoted by L1△L2, is defined in the following way. [Mursalin Sir's Past Exam Question],

L1△L2 = {w | w is in exactly one of L1 and L2}

Let Σ = {0, 1}. Consider the following languages over Σ.

A = {w | the length of w is greater than or equal to 3 but less than or equal to 5}
B = {w | the length of w is greater than or equal to 2 but less than or equal to 4}
C = {w | the length of w is odd}

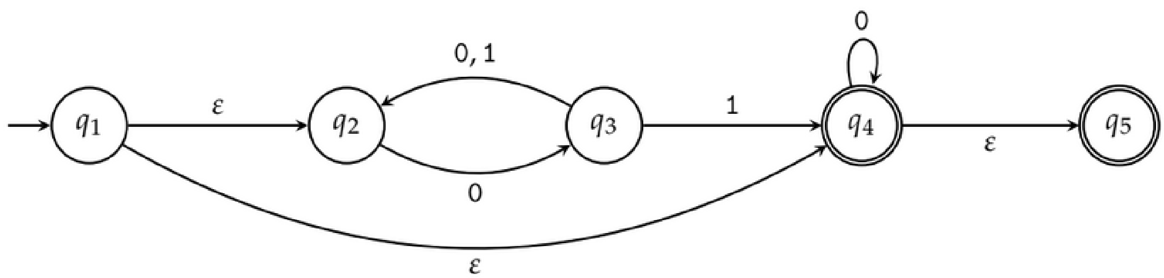
(a) Give the state diagram for a DFA that recognizes A. (2 marks)
(b) Give the state diagram for a DFA that recognizes B. (2 marks)
(c) Give the state diagram for a DFA that recognizes A△B. (2 marks)
(d) If you use the construction from class to get a DFA for the language (A△B)∪ C, how many states will it have? (1 mark)
(e) Give a 5-state DFA that recognizes (A△B) ∪ C. (3 marks)

3. **[10 Marks]** Consider the following RL over Σ = {0, 1}*

L1 = {w | w = 0^m where m is even}
L2 = {w | w = 1^n where m is even}
L3 = L1 U L2
L4 = L1 ∩ L2
L5 = L1 ⊕ L2

(a) Draw the DFA for L1.
(b) Draw the DFA for L2.
(c) Draw the DFA for L3 using cross product construction.
(d) What are the rejecting states for L4.
(e) What are the accepting states for L5.

4.**[10 Marks]** Considering the following NFA



(a) If you were to draw the equivalent DFA of the above NFA using subset construction method, what is the maximum possible rejected state? [1 mark]
(b) What is the ε- closure of q1 state in the above NFA? [1 mark]
(c) Draw the equivalent DFA of the above NFA using subset construction method. [8 Marks]

5. **[10 Marks]** Convert the following regular expressions to NFAs with ε- transitions.
(a) (0 + 1) 1* 0
(b) 0 (1 0)* 1*
(c) (0 + 1)* 1 1 (0 + 1)*
(d) (0 + (1 0))* 1
(e) ε + 0 (0 + 1)* 1

6. **[10 Marks]** Convert the following DFA to an equivalent regular expression using state elimination method. First eliminate q2, q4, finally q3. You must show working.